

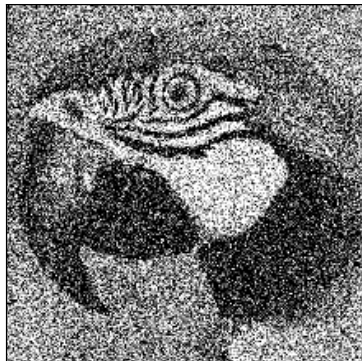
AI-powered Denoising: Unsupervised Learning

The denoising problem

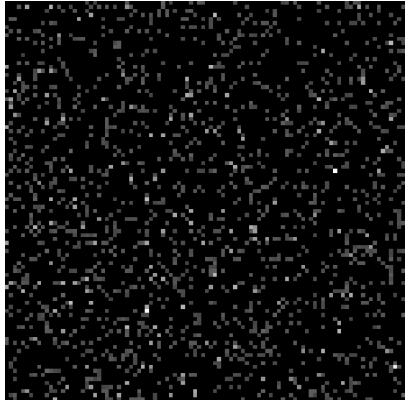
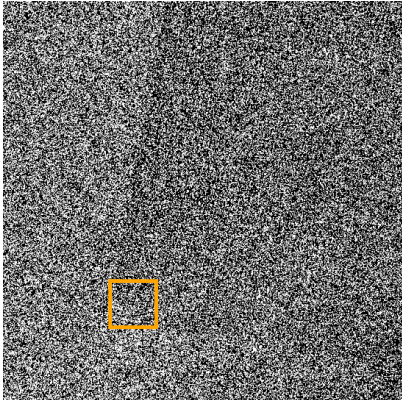
Estimate this



From this



Why?



Supervised denoising

Minimize mean squared error (MSE) on **clean** - **noisy** pairs

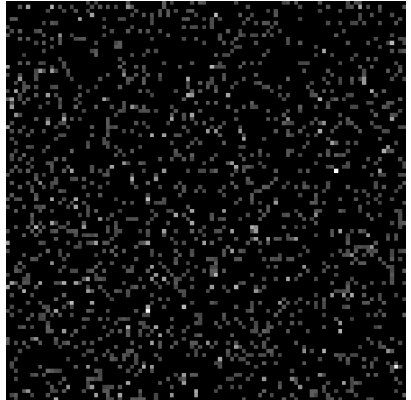
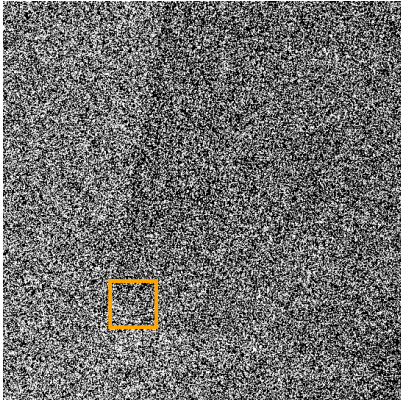
$$\left\| \begin{array}{c} \text{denoised} \end{array} - \begin{array}{c} \text{clean} \end{array} \right\|_2^2$$

The diagram illustrates the Mean Squared Error (MSE) calculation for supervised denoising. It shows two grayscale images side-by-side, separated by a minus sign. The left image is labeled 'denoised' and the right image is labeled 'clean'. Both images depict a man in a dark coat operating a large camera on a tripod in an outdoor setting. The 'denoised' image is visibly grainier than the 'clean' image. The entire expression is enclosed in a large pair of vertical bars, with a superscript '2' to the right, representing the squared L2 norm.

Good news: Works well!

Bad news: Requires clean images...

... which are often not available in practice



Plan

The Noise2Noise framework

Unsupervised methods

Real data

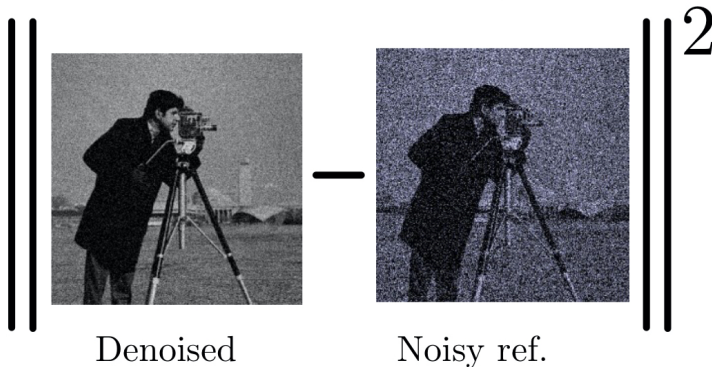
The Noise2Noise framework

Unsupervised methods

Real data

Unsupervised learning

Main insight: Learn to approximate **noisy target**, instead of clean image



Noise2Noise: Learning image restoration without clean data. Lehtinen et al. 2018

Additive noise

Clean signal: x Data: $y = x + z$ Denoised estimate: $\mathcal{D}(y)$

$$\text{MSE} := \frac{1}{n} \sum_{i=1}^n (x_i - \mathcal{D}(y)_i)^2$$

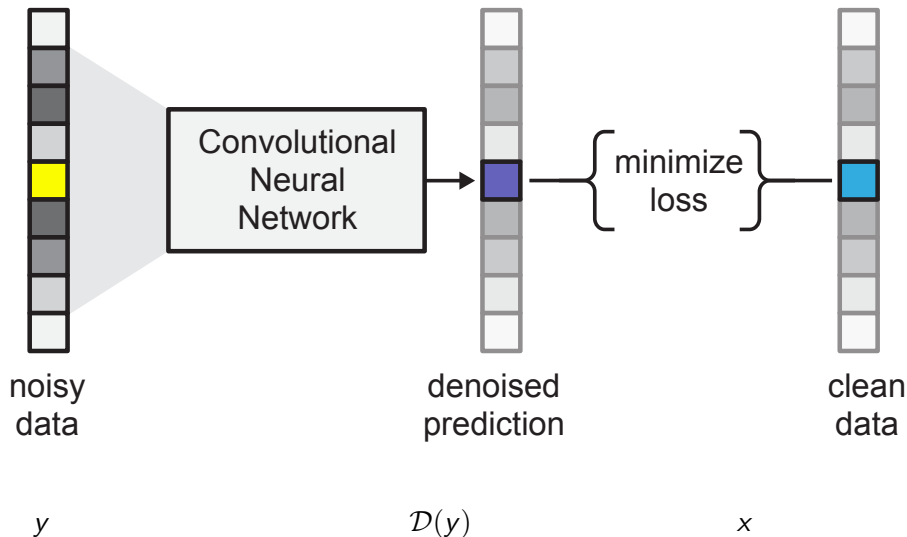
Noisy target: $t = x + w$

$$\begin{aligned}(t_i - \mathcal{D}(y)_i)^2 &= (x_i + w_i - \mathcal{D}(y)_i)^2 \\ &= (x_i - \mathcal{D}(y)_i)^2 + w_i^2 + 2w_i(x_i - \mathcal{D}(x + z))\end{aligned}$$

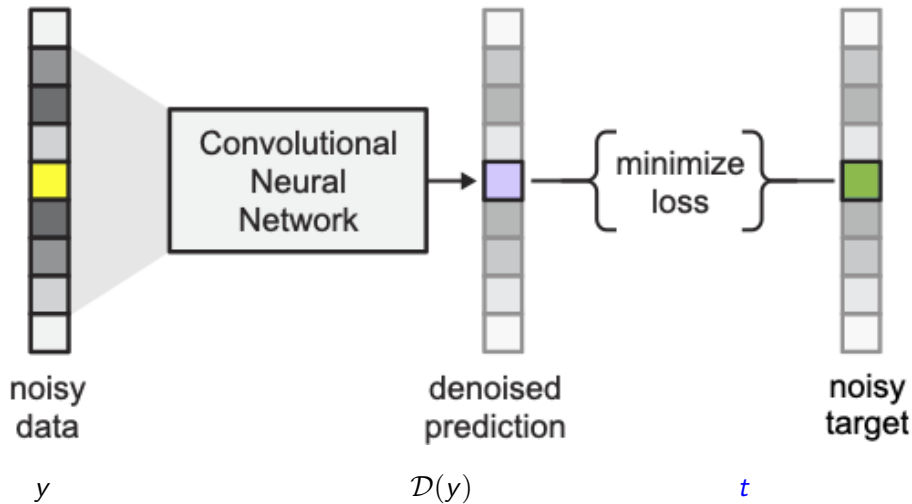
If w and z are **independent**

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n (t_i - \mathcal{D}(y)_i)^2 &\approx \frac{1}{n} \sum_{i=1}^n (x_i - \mathcal{D}(y)_i)^2 + \frac{1}{n} \sum_{i=1}^n w_i^2 \\ &\approx \text{MSE} + \frac{1}{n} \sum_{i=1}^n w_i^2\end{aligned}$$

Supervised learning



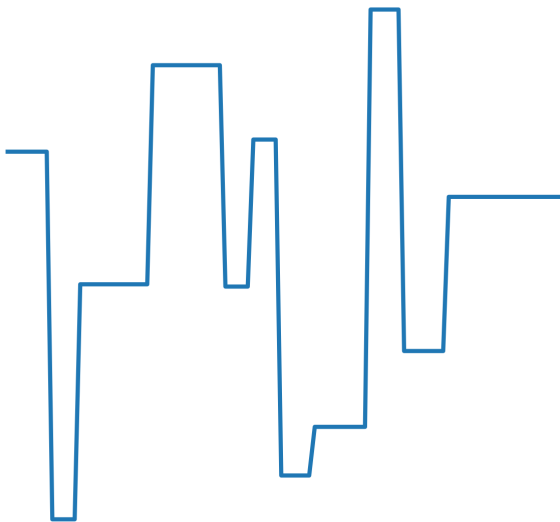
Noise2Noise



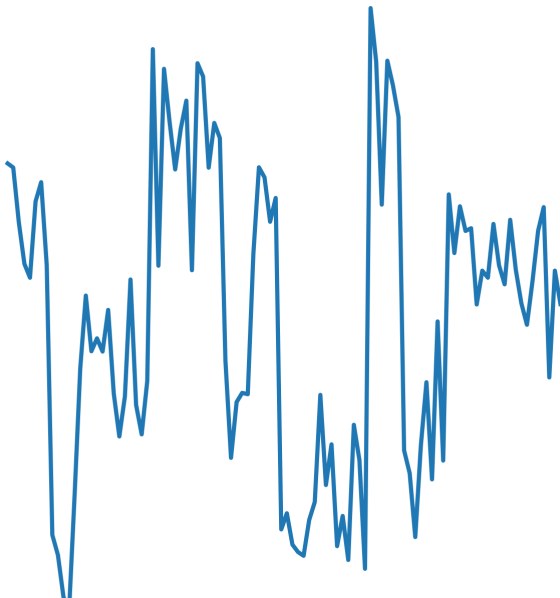
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

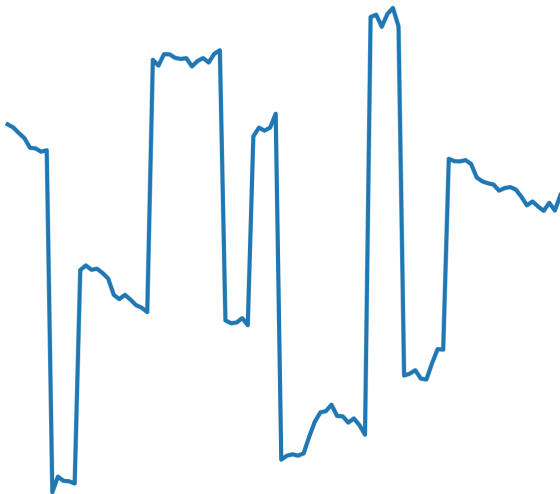
1D piecewise-constant functions



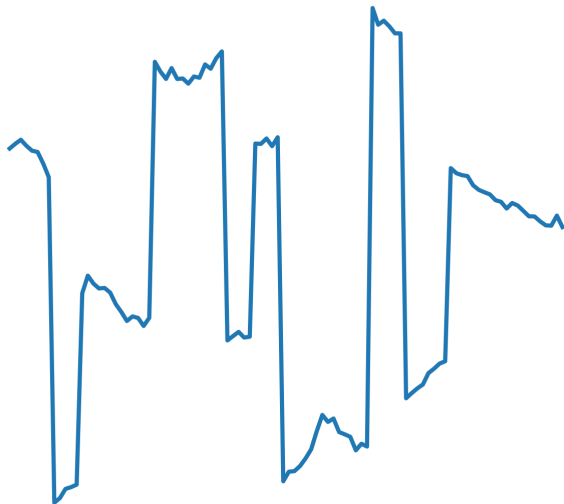
Additive i.i.d. Gaussian noise (PSNR 19.96 dB)



Supervised (PSNR 26.44 dB)



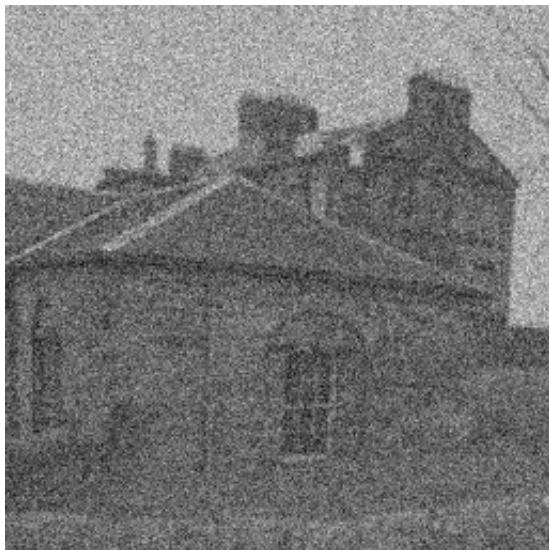
Unsupervised (PSNR 26.30 dB)



Natural images



Additive i.i.d. Gaussian noise with variable variance (PSNR 19.99 dB)



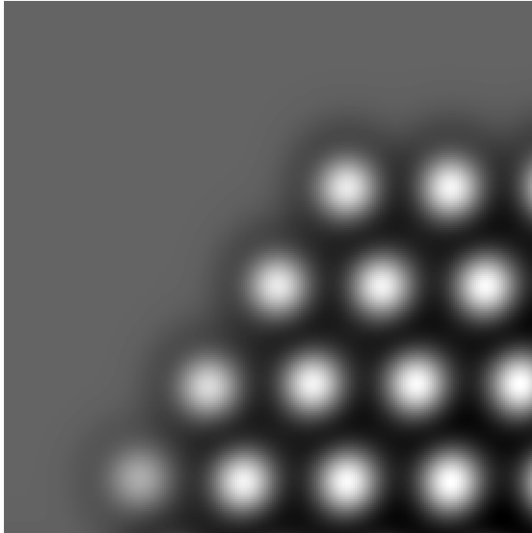
Supervised (PSNR 30.78 dB)



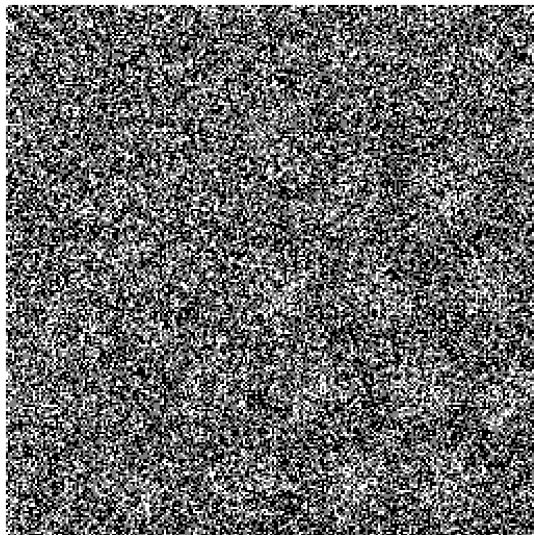
Unsupervised (PSNR 30.03 dB)



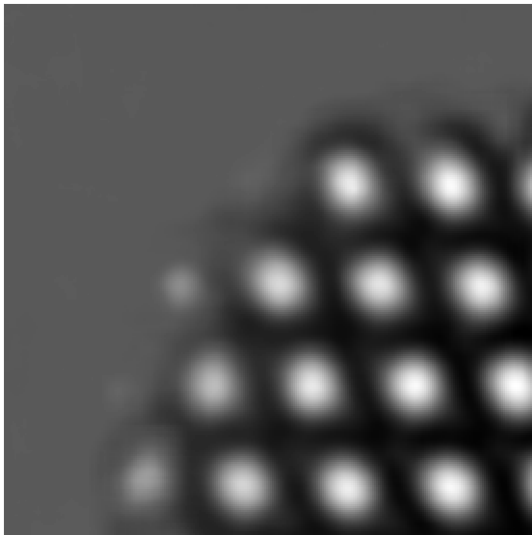
Electron microscopy



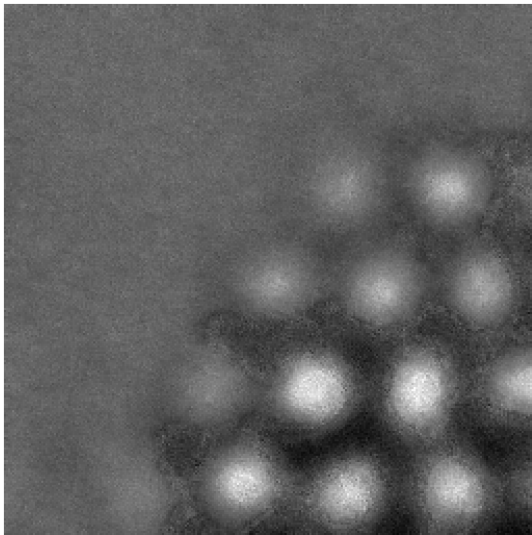
Poisson shot noise (PSNR 6.64 dB)



Supervised (PSNR 34.78 dB)

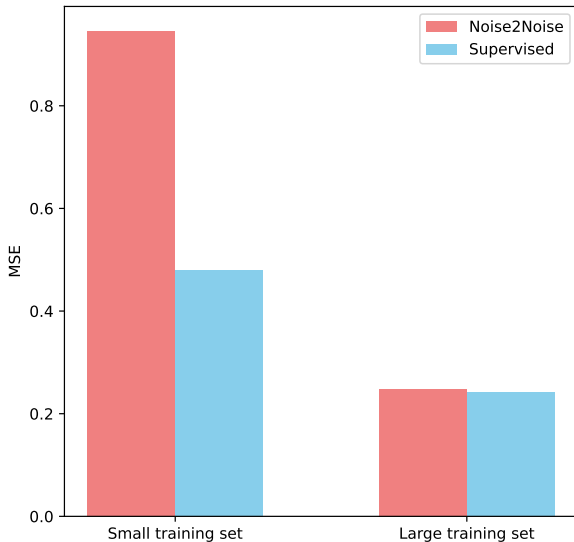


Unsupervised (PSNR 30.82 dB)



Effect of training-set size

Piecewise-constant signals



Unsupervised evaluation

Clean signal: x Data: $y = x + z$ Denoised estimate: $\mathcal{D}(y)$

$$\text{MSE} := \frac{1}{n} \sum_{i=1}^n (x_i - \mathcal{D}(y)_i)^2$$

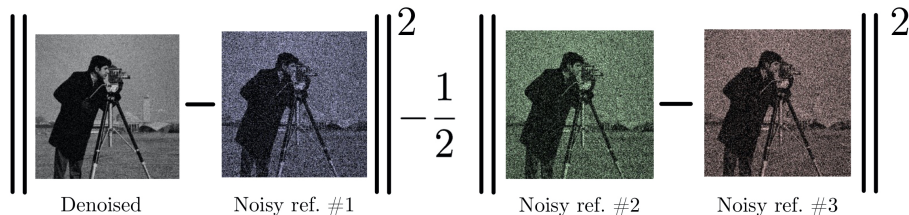
Noisy target: $t = x + w$

If w and z are **independent**

$$\frac{1}{n} \sum_{i=1}^n (t_i - \mathcal{D}(y)_i)^2 \approx \text{MSE} + \frac{1}{n} \sum_{i=1}^n w_i^2$$

Can be corrected to *estimate the supervised MSE* and perform **unsupervised evaluation**

Unsupervised MSE



$$\text{uMSE} := \frac{1}{n} \sum_{i=1}^n (t_i - \mathcal{D}(y)_i)^2 - \frac{(b_i - c_i)^2}{2}$$

If noisy target t and references b, c correspond to *same clean image* with *independent* noise, uMSE is **unbiased** and **consistent**

Evaluating unsupervised denoising requires unsupervised metrics.

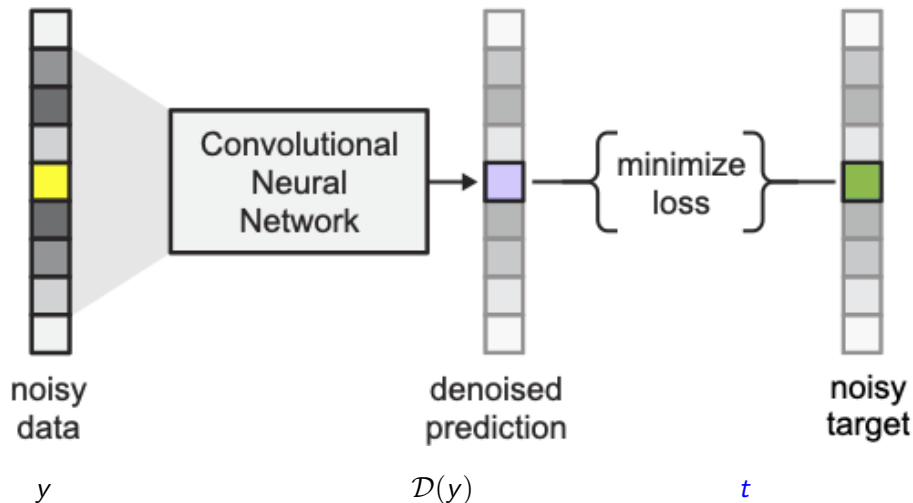
Marcos-Morales et al. 2023

The Noise2Noise framework

Unsupervised methods

Real data

Noise2Noise framework



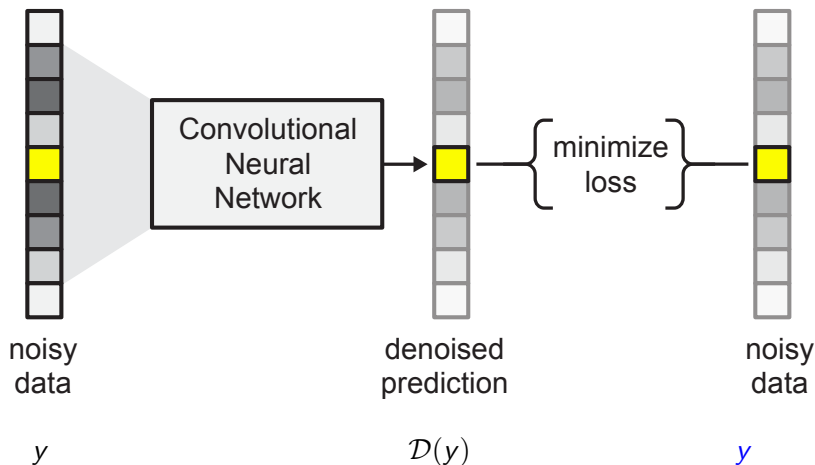
Key challenge: **How to obtain the noisy target?**

Key challenge

Ideally, target should correspond to the **same signal** as the noisy data, corrupted by **independent noise**

Different strategies to estimate the target result in different unsupervised methods

Strategy 1: Use input noisy data



Problem: Non-denoising identity function is optimal

Can work if model is **constrained** to avoid fitting noise
(*Deep image prior*. Ulyanov et al. 2018)

Strategy 2: Temporal sampling

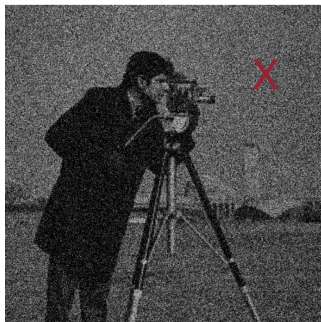
Use two frames acquired successively as noisy input and target

Assumptions:

- ▶ Noise is frame-wise independent
- ▶ Signal changes slowly over time

Strategy 3: Spatial sampling

Obtain noisy data and target via **spatial subsampling**



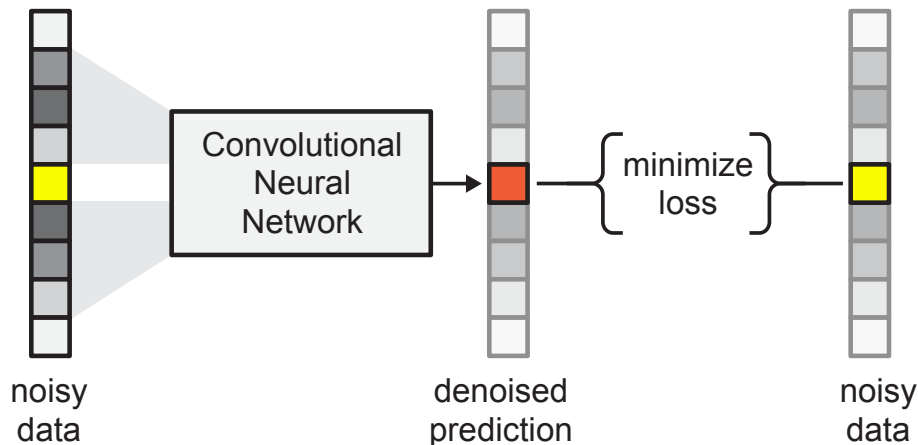
Assumptions:

- ▶ Noise is pixel-wise independent
- ▶ Signal is spatially smooth

Neighbor2Neighbor: Self-supervised denoising from single noisy images.

Huang et al. 2021

Strategy 4: Blindspot denoising



Assumptions:

- ▶ Noise is pixel-wise independent
- ▶ Signal is spatially smooth

Blindspot denoising

Two possible implementations:

1. Replace pixel by random noise in the input

Noise2Self: Blind denoising by self-supervision. Batson, Royer. 2019

Noise2Void - Learning denoising from single noisy images. Krull et al. 2019

2. Impose blindspot in the network architecture

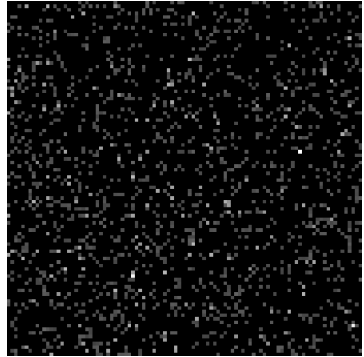
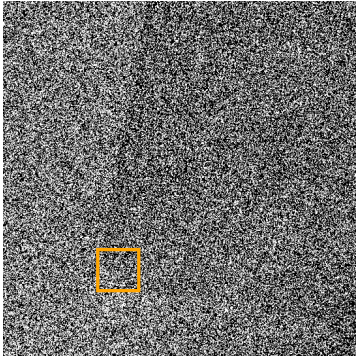
High-quality self-supervised deep image denoising. Laine et al. 2019

The Noise2Noise framework

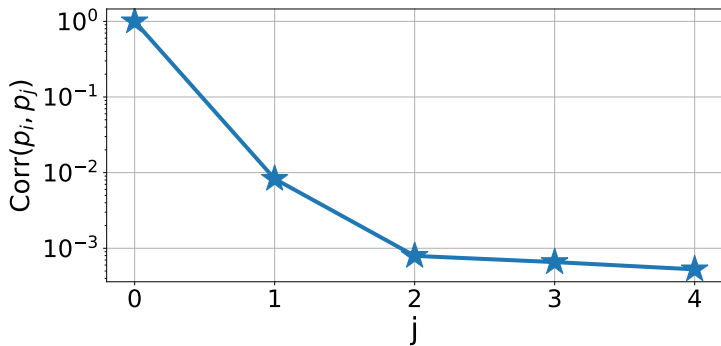
Unsupervised methods

Real data

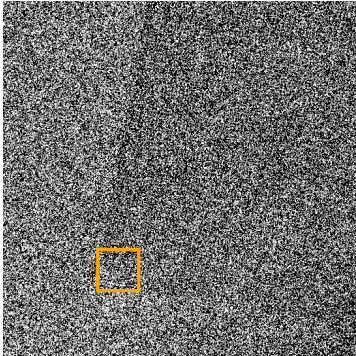
Electron microscopy



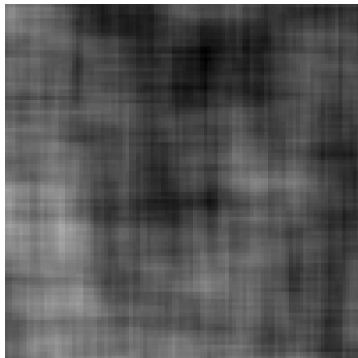
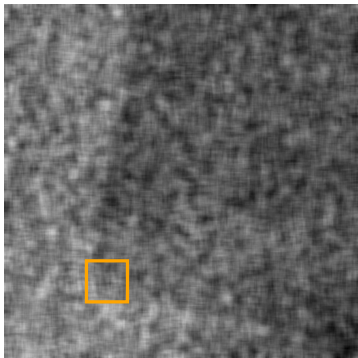
Spatial correlation of the noise



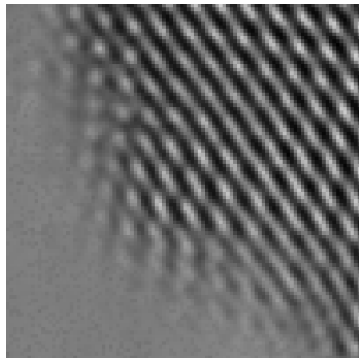
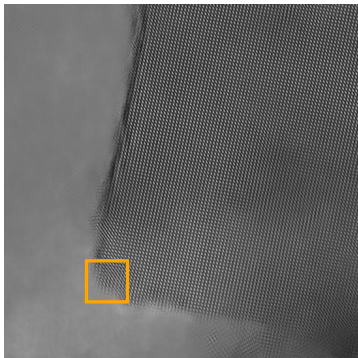
Noisy data



Gaussian smoothing, uPSNR: 20.4 dB

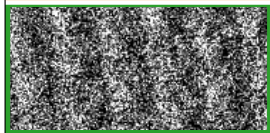
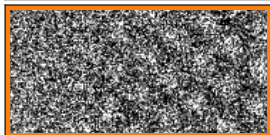
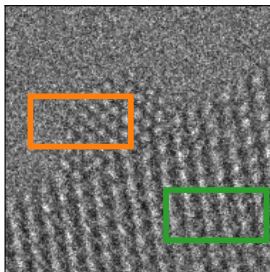


Neighbor2Neighbor, uPSNR: 26.9 dB

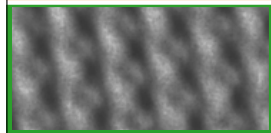
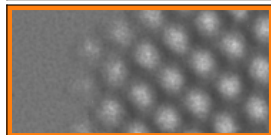
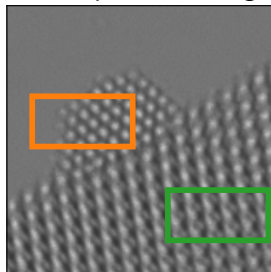


Blindspot denoising

Data



Blindspot denoising



Conclusions

We can train (and evaluate) deep denoisers without clean images!

Most existing methods are strategies to implement Noise2Noise framework

Key assumption: Noise independence

For more information

Learn, Denoise and Discover: A Guide to Deep Denoising with an Application to Electron Microscopy

Sreyas Mohan, Kangning Liu, Peter Crozier,
Carlos Fernandez-Granda

Acknowledgements

We are very grateful for the support of the [National Science Foundation](#) through grants OAC-2104105 and OAC-2103936